Dial In Number: 888-989-3317 Video: http://conf1.ls-cat.org
http://conf2.ls-cat.org

Passcode: 57615

9:00	Welcome / Approval of Minutes and Agenda (MG)
9:10	LS-CAT Status (KB)
	LS-CAT Usage
	Remote Access
	Reducing Proprietary Costs
	APS Upgrade Update
	Proposed Budget
	Budget Approval
	Executive Session (Board Members Only)
10:30 (?)	New Business / Next Meeting Planning

LS-CAT Status

Keith Brister

LS-CAT Usage Report

February 2012

Keith Brister

Manager, LS-CAT
Research Associate Professor
Synchrotron Research Center
Northwestern University

Summary

LS-CAT has been providing been providing beamtime to its members since December 2006 with basic capabilities on our 21-ID-D beamline. Since that time we've added two more experimental stations, made major improvements to our capabilities, and served nearly 900 individuals performing over 1500 experiments. This report shows a breakdown of the institutional use of LS-CAT, shows how we can make more efficient use of our resources, and suggests what must be done for the member institutions to take advantage of new opportunities at LS-CAT.

The main conclusion is that members who expect their user programs to increase in size should attempt to increase their allocation by making an arrangement with members who do not expect growth of their user programs.

Unlike many facilities, LS-CAT makes a distinction between allocated, scheduled, and used beamtime.

Hours Allocated ≠ Hours Scheduled ≠ Hours Used

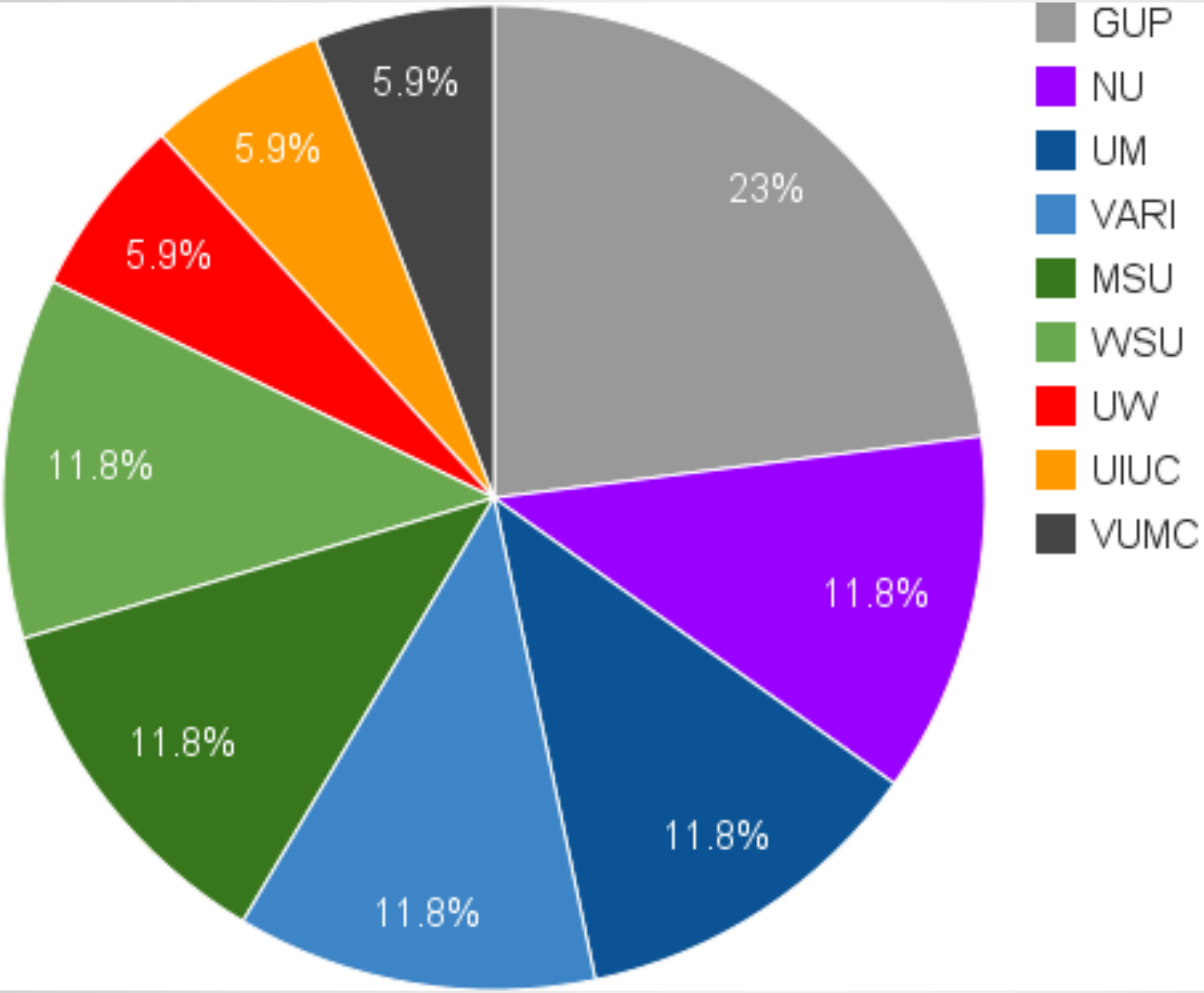
Why?

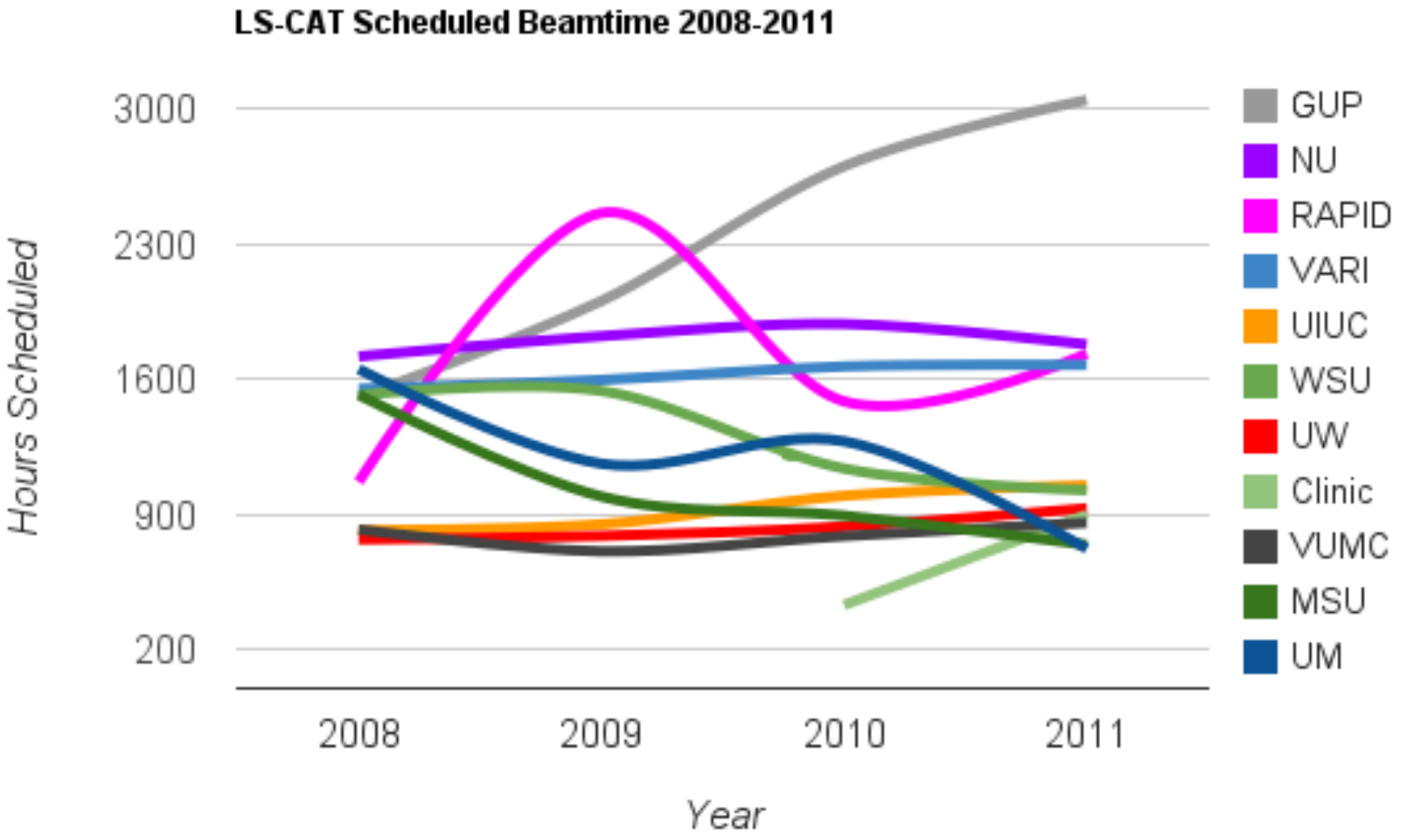
We do not want you the schedule your entire allocation if you do not need it.

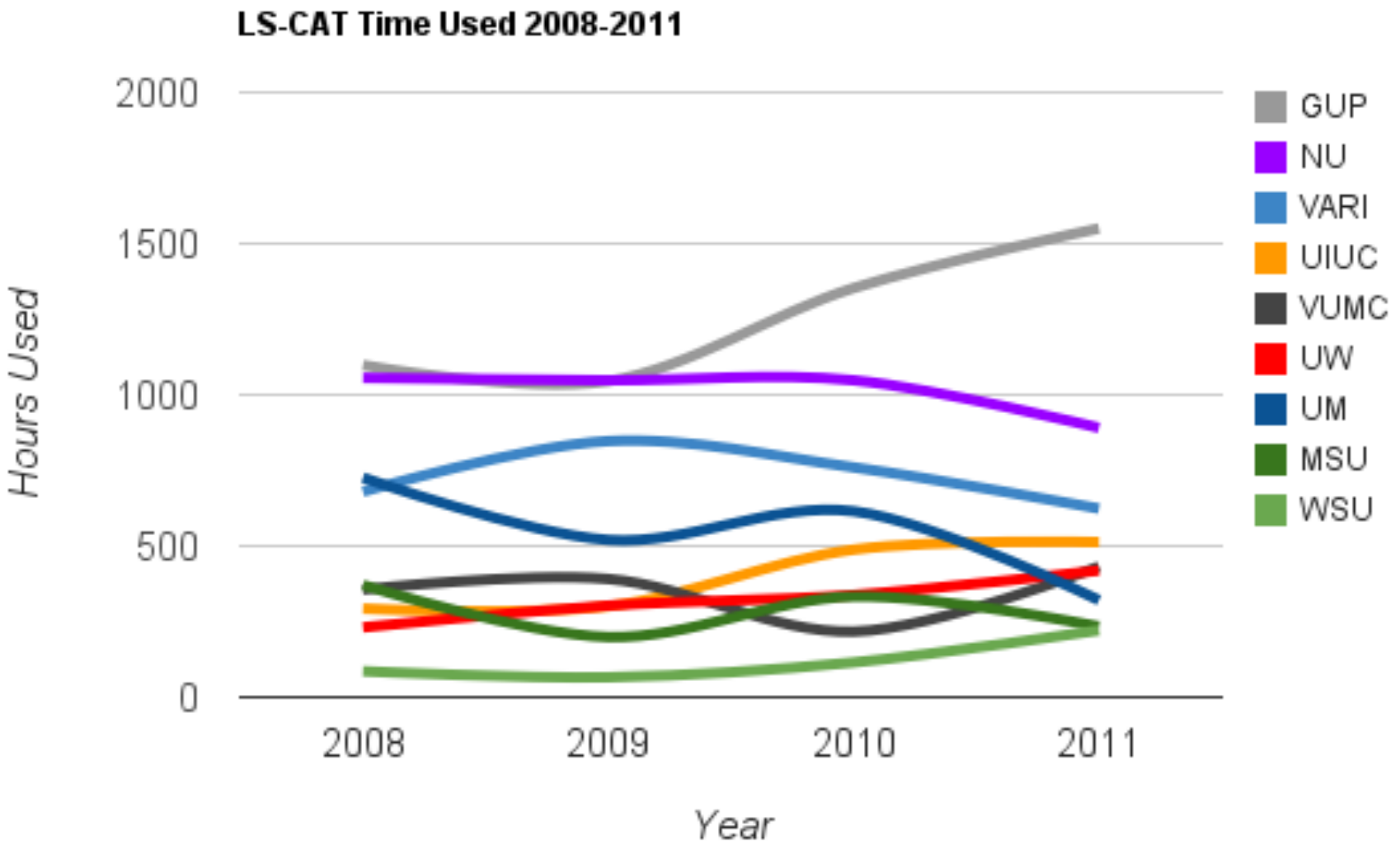
It's almost impossible to use 100% of your beamtime for collecting data: Academic users should always schedule a bit more time then they will use.

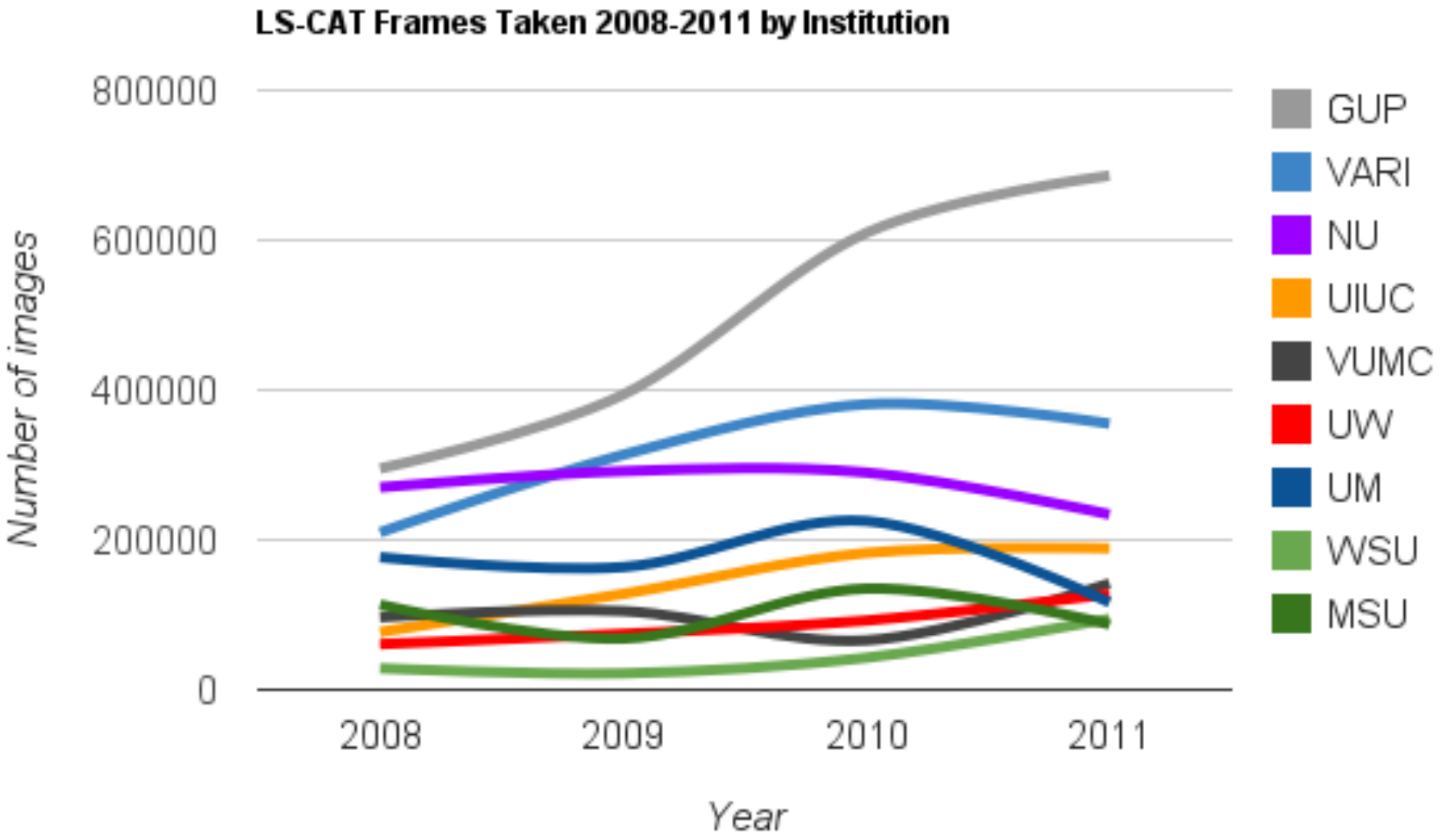
Someone else can take advantage of your unused beamtime to collect at a time they were not scheduled.

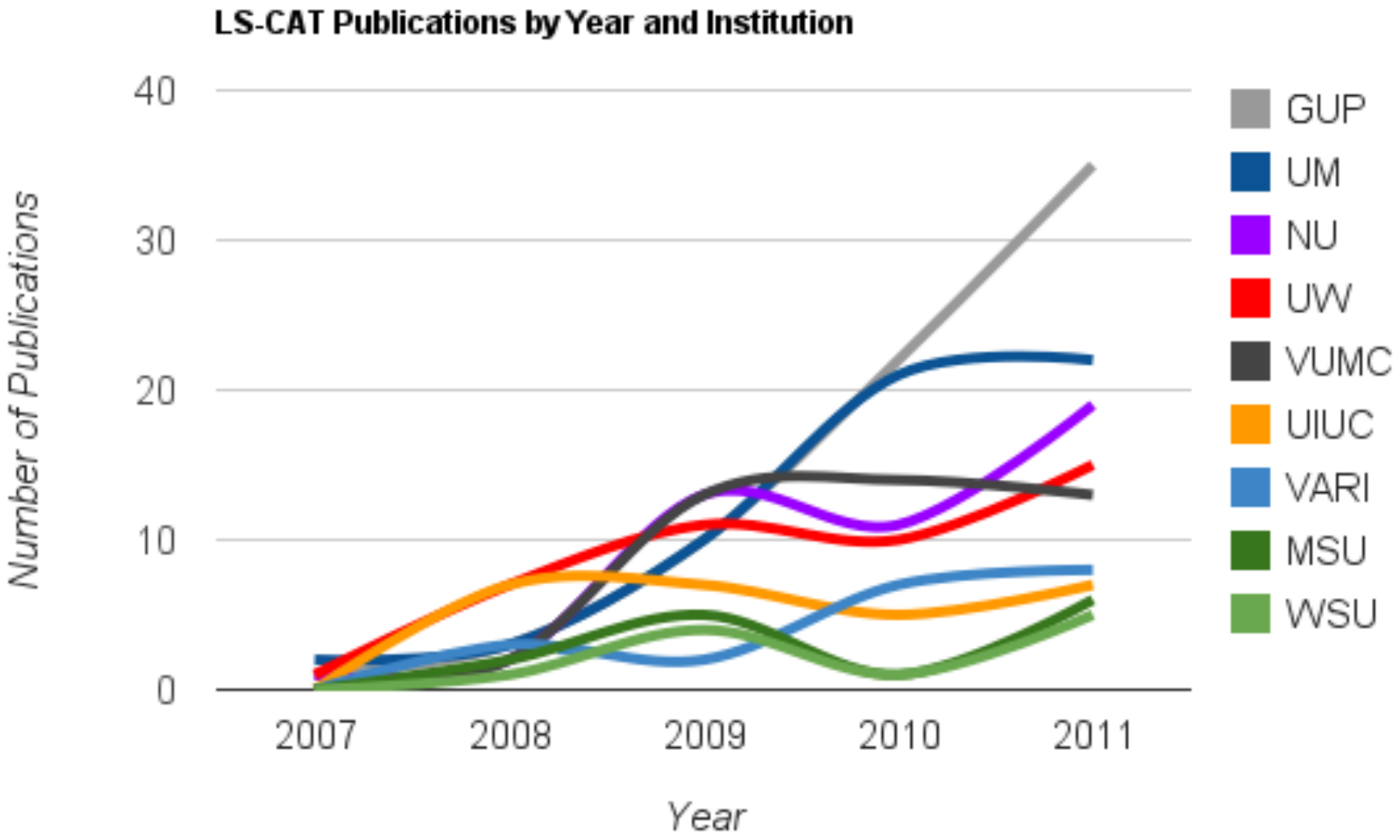
LS-CAT Beamtime Allocations

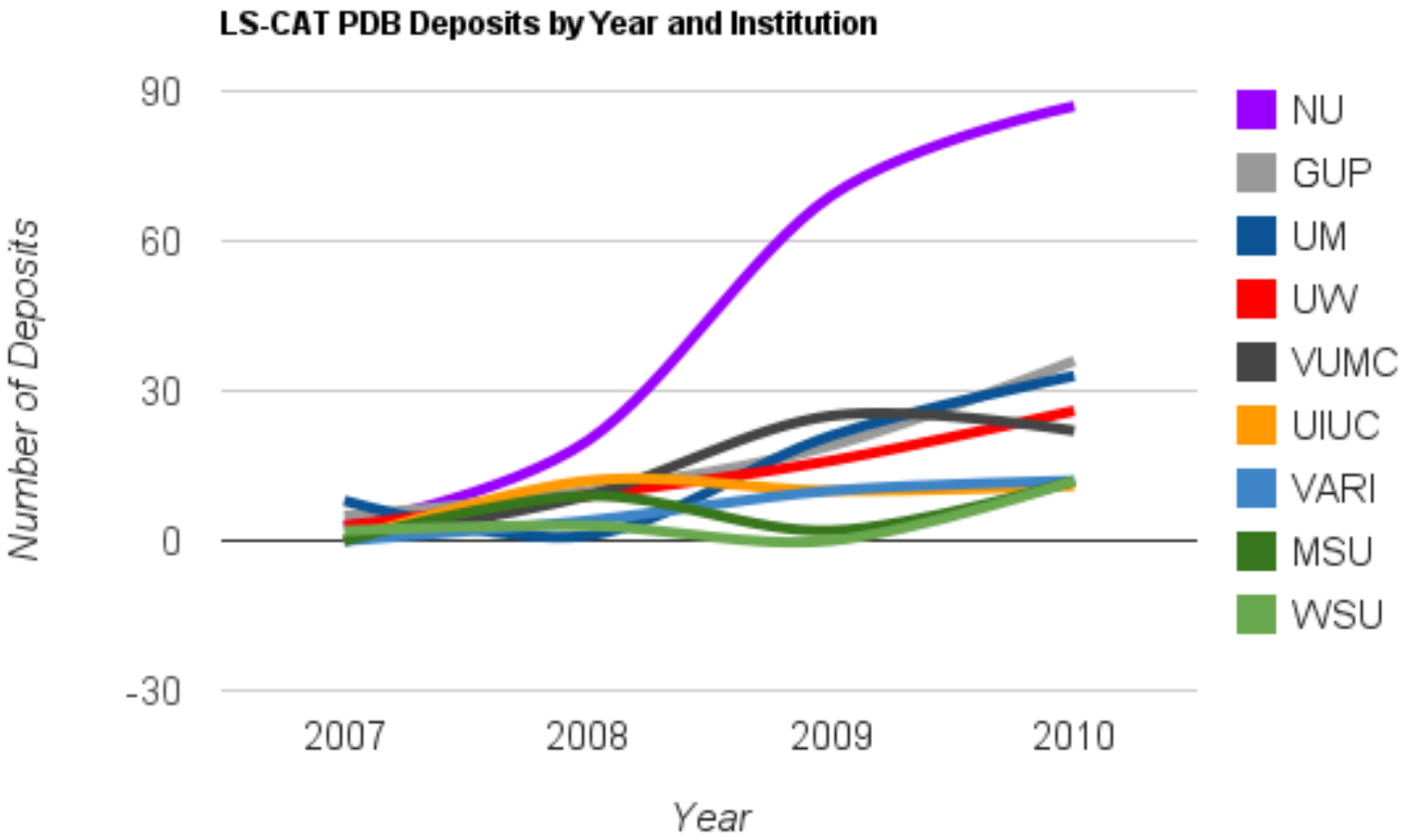


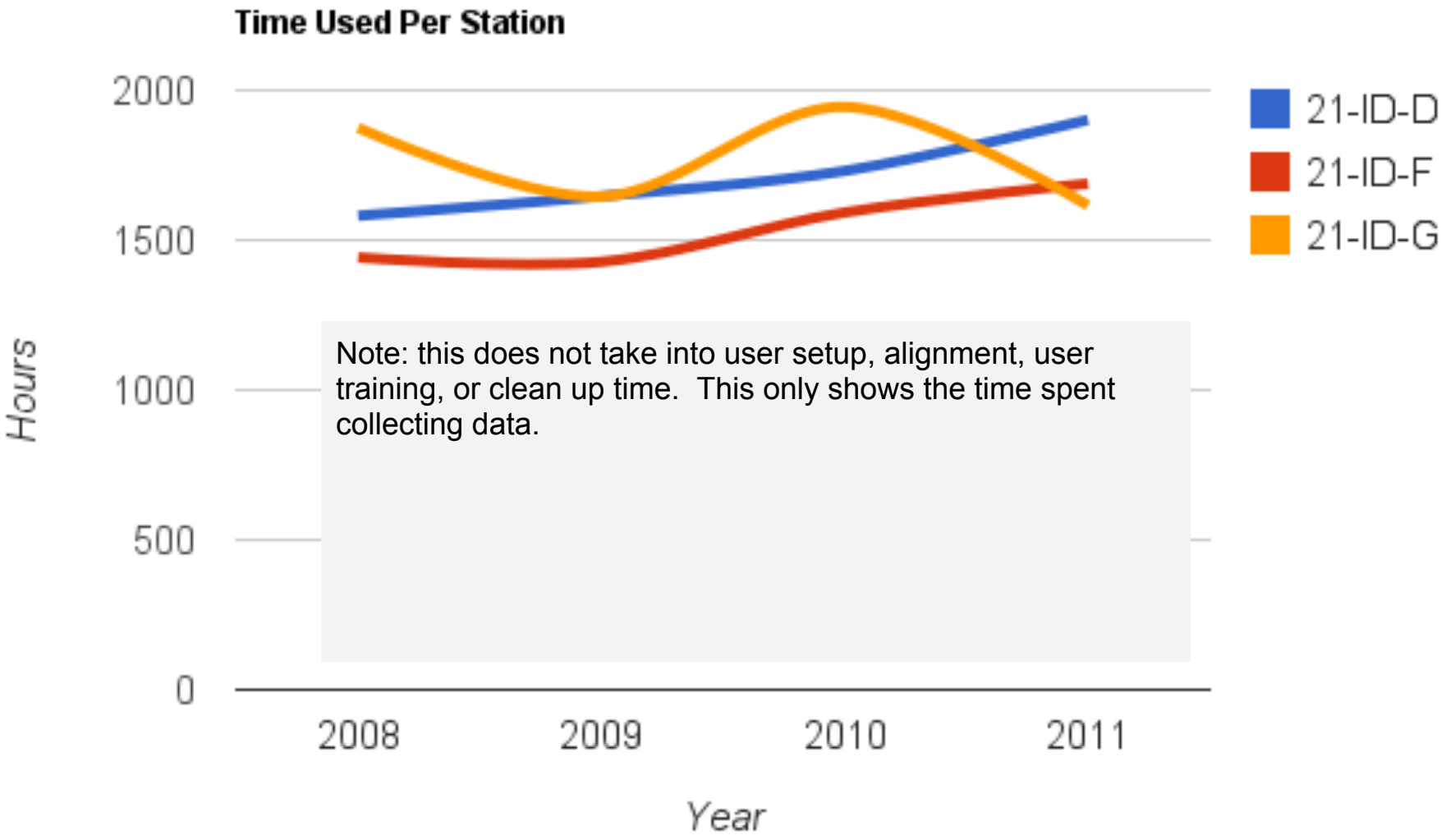

LS-CAT Scheduled Beamtime 2008-2011
Hours Scheduled
3000
2300
1600
900
200
2008
2009
2010
2011
Year
GUP
NU
RAPID
VARI
UIUC
WSU
UW
Clinic
VUMC
MSU
UM

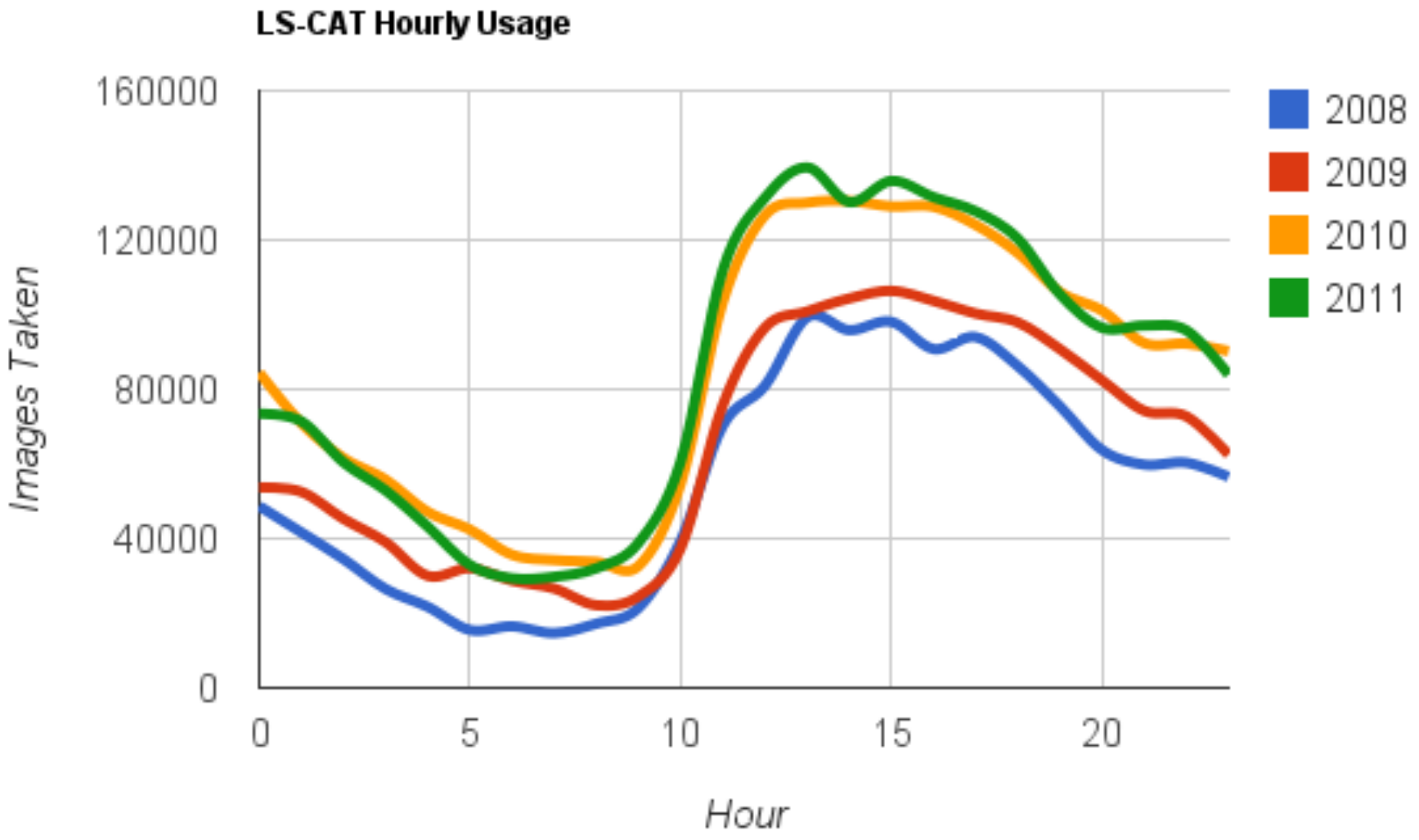

LS-CAT Time Used 2008-2011
2000
1500
1000
500
0
Hours Used
2008
2009
2010
2011
Year
GUP
NU
VARI
UIUC
VUMC
UW
UM
MSU
WSU

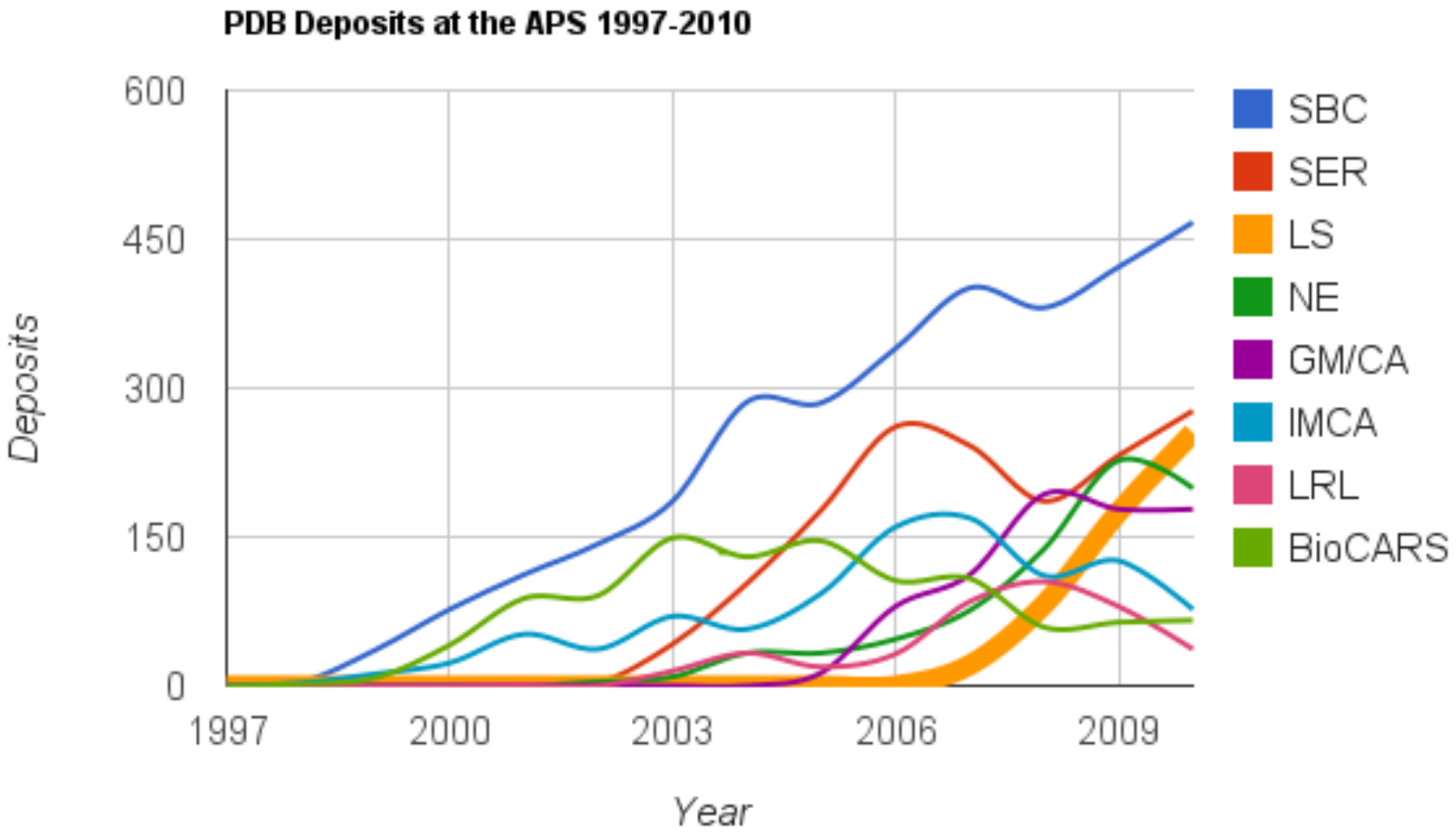

LS-CAT Frames Taken 2008-2011 by Institution
800000
600000
400000
200000
0
Number of images
2008
2009
2010
2011
Year
GUP
VARI
NU
UIUC
VUMC
UW
UM
WSU
MSU


LS-CAT Publications by Year and Institution
Number of Publications
40
30
20
10
0
2007
2008
2009
2010
2011
Year
GUP
UM
NU
UW
VUMC
UIUC
VARI
MSU
WSU


LS-CAT PDB Deposits by Year and Institution
90
60
30
0
-30
Number of Deposits
2007
2008
2009
2010
Year
NU
GUP
UM
UW
VUMC
UIUC
VARI
MSU
WSU


Time Used Per Station
2000
1500
1000
500
0
Hours
2008
2009
2010
2011
Year
21-ID-D
21-ID-F
21-ID-G
Note: this does not take into user setup, alignment, user training, or clean up time. This only shows the time spent collecting data.


LS-CAT Hourly Usage
160000
120000
80000
40000
0
Images Taken
0
5
10
15
20
Hour
2008
2009
2010
2011


PDB Deposits at the APS 1997-2010
600
450
300
150
0
Deposits
1997
2000
2003
2006
2009
Year
SBC
SER
LS
NE
GM/CA
IMCA
LRL
BioCARS

Conclusions

- It should be possible to make better use of the beamlines:
  - Run mainly remote access with short notice to users
    - Eliminate travel logistics
    - Keep all stations running whenever staff is here
  - Schedule more late afternoon and evening starts
- Some institutions are really underutilizing their allocation.

- Institutions that would like to schedule more beamtime should make arrangements to "buy" some underutilized allocation.

Remote Access

This last run (2012-1) we were caught by surprise when several groups expected 24 hour remote data collection on one or more stations without pre-arranging it or providing their own personnel as had been the custom. Further, the LS-CAT staff was often expected to know how the internal institutional time was divided up when no such schedule was provided.

New for 2012-2:

Some groups will be assumed at the time of scheduling to be running remotely and placed into times of our convenience to mesh more closely with LS-CAT personnel normal hours.

The people named on the ESAF are expected to talk to each other and coordinate beamtime internally. The spokesperson will be called upon to resolve disputes. (Independent groups within a given institution should consider submitting their own ESAF instead of joining a big ESAF.)

Remote data collection after 10PM should not be automatically assumed.

The LS-CAT staff providing support may be a technician instead of a PhD crystallographer.

Reverse Remote Access

Several other APS CATS have noticed a great lack of even basic training of users in crystallography once remote access is fully adopted. This sets up the need for "Reverse Remote Access" where the beamline staff goes to each institution to provide training in the use of the software and basic crystallography, ie, how to tell when you are collecting junk.

(Budget being proposed does not include money for Reverse Remote Access in FY 2013)

There have been discussions of having an APS or Life Sciences Council sponsored school to train graduate students and post-docs. We might consider this on a smaller scale for LS-CAT members if the larger scale efforts come to naught.

Making Data Collection Cheaper: Motivation

Note from GSK's Bob Nolte to Eric Xu:

Eric,
As I mentioned during our recent conversation, budget reductions are putting significant pressure on our synchrotron spending and will continue to do so for the foreseeable future. With high speed data collection available to GSK in the UK at Diamond and at the ESRF, the costs of collecting at the APS are disproportionally more expensive in the US. We need to address this If we hope to have any long term future at LS-CAT. Can VARI please raise the issue at the next board LS-CAT board meeting that there is an urgent need to make data collection time as efficient as possible in order for us to reduce proprietary charges. From our perspective, this is the highest priority we have currently for changes at the beam line.

This would include as many of the following as possible

Reduce dead time between frames to a minimum. Currently this is 3 seconds. If this was reduced by even 1 second we could collect 25% more data in the same period of time.

Faster detectors - obvious but expensive I am aware

Increased flux on F & G

Reduced cycle time of the auto samplers

Thanks for bringing this up with the board.

Bob Nolte

Making Data Collection Cheaper: Engineering Solutions

Reduce beamline overhead:

- Project has been moved off the back burner
- Modify the motion programs for the MD2 PMAC (0.5 sec/image ?)
- Replace the Visual Basic data collection interface (0.5 sec/image ?)

Purchase a Faster Detector:

- NIH proposal very unlikely to be funded
- We could use our entire capital reserve + something to purchase a faster detector

Increase Flux on F and G:

- White beam CRL in 21-ID-A to collimate beam (possible factor of 2 gain in flux on sample)
- NE-CAT plans to do something similar have been abandoned
- Present JJ-Xray design would require significant modification
- Likely run $100K – $200K, closer to $100K if we provide the modifications to the JJ design.

Making Data Collection Cheaper: Political Solutions

Address APS proprietary rate formulation:

- Current rate is Total cost (including all of XSD) / Total number of operating ports.
- APS is very reluctant to open this can of worms
- Current administration is less reluctant than the previous one
- Best case scenario: reduces rate by factor of 4
- Worst case scenario: increases rate by factor of 2

Horse trade with APS for faster detector:

- APS BNP group wants more time: A faster detector would allow the same amount of data to be collected on D in fewer days. Would likely look like an increase in the amount of GU time on D.

- APS Upgrade opportunities (see next slide)

APS Upgrade Update

Program	BASE	Option 1	Option 2
	J'	O	J
DCS	35-ID	35-ID	35-ID
ISN	32-ID-2	9-ID-2	9-ID-2
MS-S	2-ID	2-ID	2-ID
RIXS	27-ID	27-ID	27-ID
S3DD micro	34-ID-2	34-ID-2	34-ID-2
S3DD nano	34-ID-1s	34-ID-1s	34-ID-1s
WFI	20-ID	20-ID	20-ID
XIS	28-ID	28-ID	28-ID
ASD-D	30-ID-s	30-ID-s	30-ID-s
ASL	25-ID	25-ID	25-ID
BCDI	9-ID-2	9-ID-1	32-ID-2
BIO	18-ID	18-ID	18-ID
MD	2-ID-1	32-ID-2	2-ID-1
SBC	19-ID	19-ID	19-ID
XSD LSS	28-ID*	28-ID*	28-ID*
XSD mDiff	34-ID-1s	34-ID-1s	34-ID-1s
XSD mFluor	9-ID-1	2-ID-1	9-ID-1

<=== The choice 20-ID for the Wide Field Imaging beamline puts a beam pipe through our lab.

Unfortunately, the APS views the alternatives as expensive and risky.

Fortunately, we can negotiate compensation at a time when the APS has a bit more money than normal. However, It is highly unlikely that anything as expensive as a detector will be extracted from the upgrade project, so don't get your hopes up.

Representing Life Sciences in the upgrade are:

Keith Moffat, Senior Life Sciences Advisor

Robert Fischetti, Associate Division Director

Chris Jacobson, Associate Division Director

Keith Brister, Life Sciences Council Chair

LS-CAT FY13 Proposed Budget (Detail)

Code	Description	FY2012 Total (Est)	FY2012 Budget	FY2013 Proposed	Difference
	Salaries	776,818	804,558	847,424	42,866
60180	Fringe Benefits	209,383	204,202	228,488	24,286
73000	Supplies	74,135	160,000	76,000	-84,000
73001	Supplies, Restricted	6,514	5,000	5,000	0
75000	Services	65,919	35,000	66,000	31,000
75002	Services, Restricted	12,033	16,000	12,000	-4,000
75491	External Space Rental	106,733	106,733	109,935	3,202
76761	Travel, Domestic	4,649	14,000	2,000	-12,000
76900	Utilities and Telephone	15,836	17,000	8,000	-9,000
77500	Capital Equipment	231,003	175,000	208,000	33,000
78710	Indirect Cost - Sponsored	283,703	412,507	387,153	-25,354

Estimated capital fund at start of FY13: $1.6M

This proposed budget is in line with current spending: Increases in the payroll are made up by an overall decrease in budgeted spending coupled with a decrease in the amount set aside for new detectors. Balancing the budget without increasing the dues or dipping into the capital fund will work for another 1 to 2 years, depending on the assumptions one makes.

LS-CAT FY 2013 Proposed Budget (2 Votes Required)

Salaries				
	Base	847,424		
	Fringe	228,488		
	Total		1,075,912	
Items without Indirect Costs				
77500	Capital Equip.	208,000		
75491	Rent	109,935		
76900	Telecom	8,000		
	Total		325,935	
Non-Salary Items with Indirect Costs				
73000	Lab Supplies	76,000		
73001	Office Supplies	5,000		
75000	Lab Services	66,000		
75002	Insurance	12,000		
76761	Travel	2,000		
	Total		161,000	
78710	Indirect Costs		387,153	
Total			**1,950,000**	
Per Share			**300,000**	<===== ANL, MSU, VARI, WSU
Per 0.5 Share			**150,000**	<===== UIUC, VU
			120,000	<====== UW
			30,000	<====== MCW

100% KB, NB, JA, MB, JVO, EK, DS, SA
90% ZW, JSB

Vote 1: Capitalization of maintenance when possible.

Pre-approval is requested to use the capital equipment line for purchase of replacement equipment required to maintain the facility.

Note:
Salaries are only in the planning stage at this point. Any increase or decrease between the planned salaries and the actual salaries will be made up using the capital equipment line and will not change the amount billed.

Bills are due and payable September 1, 2012. Payment arrangements with NU must be made by September 30, 2012 to avoid being declared delinquent.

Vote 2: Amounts